

DeepSentry

Unsupervised Anomaly Detection in Distributed System Logs using Autoencoders.

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Abstract

- We are engineering an intelligent cybersecurity tool that autonomously detects malicious activity in server logs by learning the normal baseline behavior of the system without needing labeled attack data.
- We start by preprocessing unstructured log data and mapping it to high dimensional vectors to capture the semantic relationships between different system events.
- To efficiently process this data we implement an Auto Encoder from Module 5 to compress the feature space and force the model to learn the most salient characteristics of normal operations.
- We then pass these compressed representations into a Recurrent Neural Network utilizing the Encoder Decoder architecture from Module 3 to predict the next expected system state based on the history of events.
- We treat any significant deviation between the predicted state and the actual state as an anomaly which triggers an alert for potential security breaches.
- Throughout the training phase we utilize Optimization Strategies from Module 1 specifically Adaptive Learning Rates to handle the sparse and noisy nature of real world system logs.

Problem Statement

- Traditional cybersecurity systems rely on rule-based signatures that can only detect known attacks.
- This leaves systems vulnerable to zero day exploits and novel intrusion methods.
- Our project addresses this gap by building a system that models the "normal" state of a distributed system.
- By learning the probability distribution of standard system logs, our model can identify any event that falls outside this distribution, effectively flagging unknown attacks without human intervention.

The Core Concept

- Instead of teaching the AI what an attack looks like (which requires millions of attack samples we do not have), we teach it what "normal" looks like.
- If the system sees something that is "not normal," it flags it.
- This approach allows us to detect entirely new forms of cyber attacks that have never been seen before because they will inherently deviate from the established baseline of normal system behavior.

Methodology & Technical Approach

1. Data Preprocessing & Dimensionality Reduction (Module 5)

- We will ingest raw server logs (e.g., HDFS or Linux system logs) and convert the text data into numerical vectors.
- Since log data is extremely high dimensional, we will apply **Dimensionality Reduction** techniques as covered in Module 5.

- Specifically, we will design an **Auto Encoder** neural network to compress the input vector into a lower dimensional latent representation.
- This ensures the model focuses on the core signals of the system rather than noise.

2. Sequence Modeling with RNNs (Module 3)

- System logs are temporal; the order of events matters.
- To capture this, we will implement a **Recurrent Neural Network (RNN)** using **Long Short Term Memory (LSTM)** units.
- We will employ the **Encoder Decoder Sequence to Sequence Architecture** where the Encoder reads the past sequence of compressed logs and the Decoder predicts the next expected log vector.

3. Optimization & Regularization (Module 1)

- To ensure our model generalizes well and does not just memorize the training data, we will apply **Regularization** techniques such as **Parameter Norm Penalties**.
- We will train the network using **Algorithms with Adaptive Learning Rates** (specifically Adam) to ensure convergence despite the sparse nature of log data.

Syllabus Compliance Matrix

- **Module 1 (Optimization Strategies):** Using Adam optimizer for sparse log data convergence.
- **Module 1 (Regularization):** Applying Dropout and L2 Norm Penalties to prevent overfitting.
- **Module 3 (Encoder Decoder Seq2Seq):** Using LSTM Encoder Decoder to model time series log patterns.
- **Module 3 (Recurrent Neural Networks):** Capturing temporal dependencies in system event logs.
- **Module 5 (Auto Encoder):** Compressing high dimensional text data into latent features.
- **Module 5 (Dimensionality Reduction):** Reducing noise in raw logs to improve detection accuracy.

Course & Program Outcome Mapping

- **CO1:** We apply advanced regularization techniques to improve the generalization of our deep learning model on unseen log data.
- **CO3:** We demonstrate an understanding of RNN basics by applying LSTM units to solve the problem of sequential anomaly detection.
- **CO5:** We utilize Auto Encoders for feature extraction and dimensionality reduction on complex datasets.
- **PO3 (Design/Development of Solutions):** We are designing a system component that directly addresses public safety and security considerations in digital infrastructure.
- **PO5 (Modern Tool Usage):** We are selecting and applying modern deep learning frameworks (PyTorch/TensorFlow) to model complex engineering activities.

Tools & Technologies

- **Language:** Python
- **Deep Learning Framework:** PyTorch or TensorFlow
- **Data Processing:** Pandas, Scikit Learn (for vectorization)
- **Visualization:** Matplotlib (for plotting reconstruction error thresholds)